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SSLC MODEL EXAMINATION 2026

MATHEMATICS – ANSWER KEY ME 531

Qn no	Key	Score
	SECTION A	
1	(c) 35	1
2	(b) (0,3)	1
3	(d) $4n+3$	1
4	(a) (2,7)	1
5	(d) 60	1
6	(b) 90	1
7	(c) (ii) and (iii) are true.	1
8	(d) Both statements are true and statement 2 is the reason of statement 1.	1
	SECTION B	
9	Probability $= \frac{30}{360} + \frac{40}{360} + \frac{50}{360}$ $= \frac{120}{360} = \frac{1}{3}$	1 1
10A	(i) Probability of getting an apple $= \frac{20}{50} = \frac{2}{5}$ (ii) Probability of getting an orange $= \frac{30}{50} = \frac{3}{5}$ (iii) Probability of getting an orange after putting 10 apples $= \frac{30}{60} = \frac{1}{2}$	1 1 1
10B	(i) $15 \times 16 = 240$ (ii) Probability of both being white $= \frac{7 \times 6}{240} = \frac{42}{240} = \frac{7}{40}$ (iii) Probability that at least one is white $= 1 -$ Probability of both being blue $= 1 - \frac{8 \times 10}{240} = \frac{160}{240} = \frac{2}{3}$ OR Probability that at least one is white $= \frac{(7 \times 6) + (7 \times 10) + (8 \times 6)}{240} = \frac{160}{240} = \frac{2}{3}$	1 1 1
11A	(i) $x_4 = \frac{140}{7} = 20$ (ii) $x_5 = 25$ $\left[x_4 + x_5 = \frac{180}{4} = 45 \right]$ OR $x_5 = 20 + 5 = 25$ $\left[x_8 = 180 - 140 = 40 \ , 4d = 20 \ , d = 5 \right]$ (iii) Sum $= 5(25 + 30) = 275$ [Sum $= 5(x_5 + x_6)$]	1 1 1
11B	(i) $\frac{15 \times 16}{2} = 120$ (ii) $x_n = 6n + 1$	1 1

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Sum $= 6 \times \frac{15 \times 16}{2} + (1 \times 15)$ [Sum $= 6 \times \frac{n(n+1)}{2} + (1 \times n)$] 1

$= (6 \times 120) + 15 = 735$ 1

12

Daily wages	Number of workers
Below 500	7
Below 600	15
Below 700	25
Below 800	34
Below 900	39
Below 1000	43

$N = 43$

(i) Median daily wage = Daily wage of the 22nd worker $= x_{22}$ 1

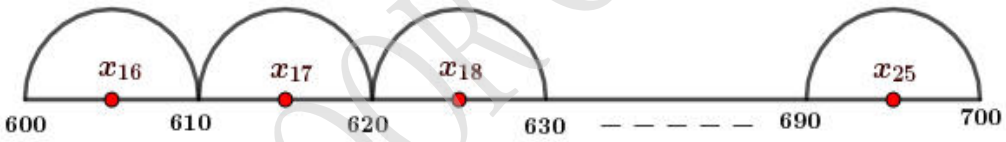
Median class $= 600 - 700$

There are 10 workers in the median class 1

Divide the 100 rupees between 600 and 700 into 10 equal parts .

Length of one sub division $= \frac{100}{10} = 10 = d$ 1

Assume that each such subdivision contains one worker whose daily wage is the mid value of that subdivision . .



(ii) Daily wage of the 16th worker $= x_{16} = \frac{600 + 610}{2} = 605\ Rs$ 1

The daily wages in the median class are in arithmetic sequence

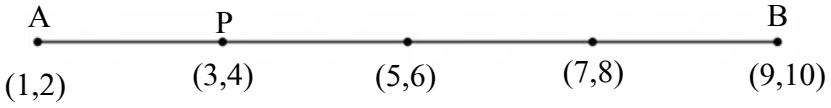
(iii) Median daily wage $= x_{22} = 605 + (6 \times 10) = 665\ Rs$ 1

SECTION C

13

(i) Coordinates of the midpoint $= \left(\frac{1+9}{2}, \frac{2+10}{2}\right) = (5, 6)$ 1

(ii)

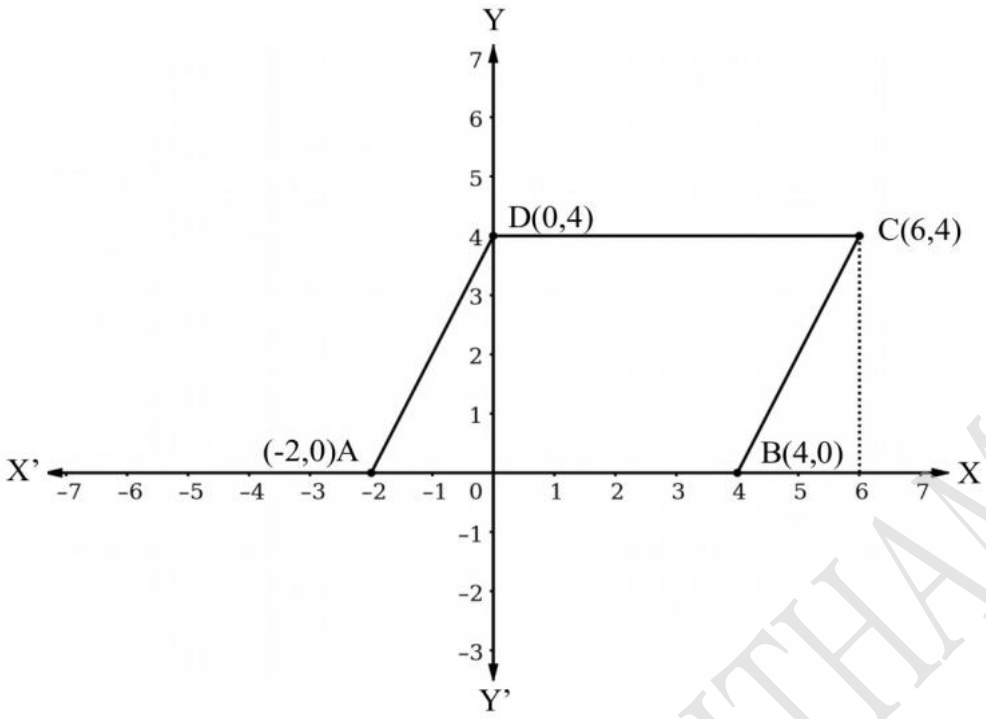


1

Coordinates of P = (3,4) 1

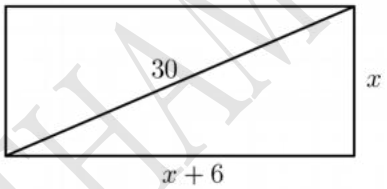
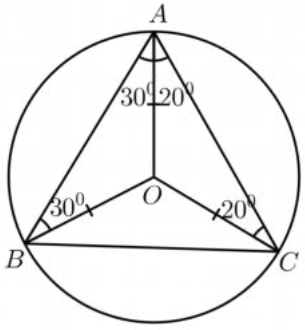
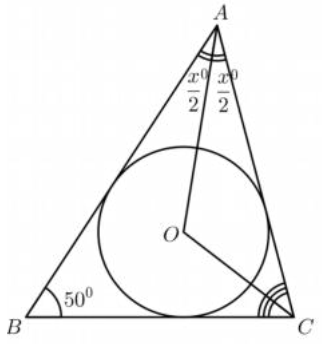
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14	(i)  (ii) Area of the parallelogram ABCD $= 6 \times 4 = 24\ sq.units$	3 1
15A	(i) Slope $= \frac{7-1}{2-(-1)} = \frac{6}{3} = 2$ (ii) $(3,9)$ OR $(5,13)$ [Any point got from the given points such that as the change in x coordinates , then the change in y is 2] (iii) $\frac{2x+3-1}{x-(-1)} = \frac{2x+2}{x+1} = \frac{2(x+1)}{x+1} = 2$ Since the slope is the same , $(x, 2x+3)$ is a point on this line. OR If (x,y) is a point on this line , $\frac{y-1}{x-(-1)} = 2 \implies \frac{y-1}{x+1} = 2 \implies y-1 = 2(x+1)$ $y-1 = 2x+2 \implies y = 2x+3$	1 1 2
15B	(i) $(10,0)$ OR $(-10,0)$ (ii) $\sqrt{(5-0)^2+(9-0)^2} = \sqrt{25+81} = \sqrt{106}$ $(5,9)$ is outside the circle. [$\sqrt{106}$ is greater than 10] (iii) $(x-0)^2+(y-0)^2 = 10^2 \implies x^2+y^2 = 100$	1 1 1 1
	SECTION D	
16	(i) 1 (ii) The squares of the numbers leave remainder 1 on division by 3 are again leave remainder 1 on division by 3 . So the squares of each term is also a term of this sequence. OR $x_n = 3n+1$ [$x_n = dn + f - d$] $x_n^2 = (3n+1)^2 = 9n^2+6n+1 = 3(3n^2+2n)+1$	1 2

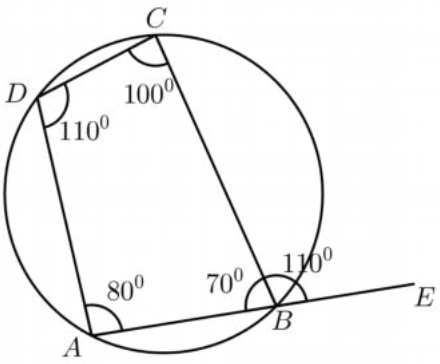
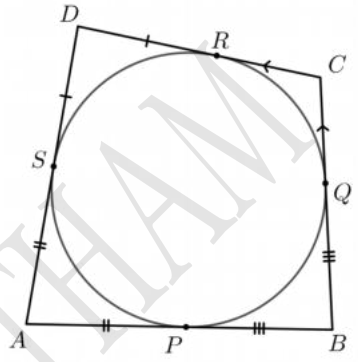
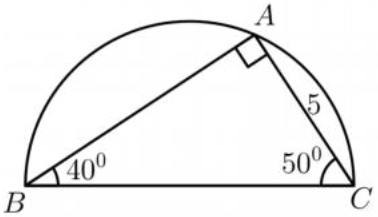
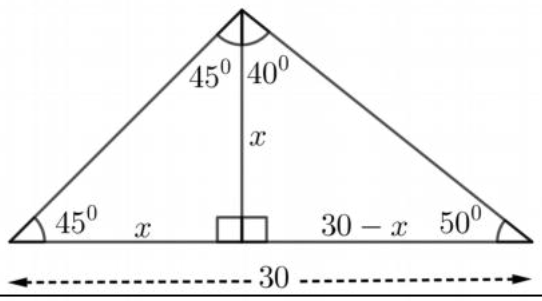
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	The squares of each term of this sequence leaves remainder 1 on division by 3 . So the squares of each term is also a term of this sequence.	
17A	(i) $3 + 6 + 9 + \ldots + 60 = 3(1 + 2 + 3 + \ldots + 20)$ $= 3 \times \frac{20 \times 21}{2} = 630$ (ii) $6 + 12 + 18 + \ldots + 120 = 2(3 + 6 + 9 + \ldots + 60) = 2 \times 630 = 1260$ (iii) $1260 + (20 \times 2) = 1300$	1 1 1 1
17B	(i) $x_2 + x_{29} = 60$ $\quad [\, x_1 + x_{30} = x_2 + x_{29} \,]$ (ii) $x_{15} + x_{16} = 60$ $\quad [\, x_1 + x_{30} = x_2 + x_{29} = x_{15} + x_{16} \,]$ (iii) Sum $= 15 \times 60 = 900$	1 1 1
18	(i) Length of the shorter side = x $==>$ Length of the larger side = x + 6 $(x+6)^2 + x^2 = 30^2 \quad ==> \quad 2x^2 + 12x + 36 = 900$ (ii) $2x^2 + 12x = 900 - 36 = 864$ $x^2 + 6x = 432$ $x^2 + 6x + 3^2 = 432 + 3^2$ $(x+3)^2 = 441$ $x + 3 = \sqrt{441} = 21 \quad ==> \quad x = 18$ Lengths of the sides = 18 cm. , 24 cm. 	1 1 1 1
19	(i) $3 \times 5^2 + (6 \times 5) = 75 + 30 = 105$ (ii) $3n^2 + 6n = 360 \quad ==> \quad 3(n^2 + 2n) = 3 \times 120 \quad ==> \quad n^2 + 2n = 120$ $n^2 + 2n + 1^2 = 120 + 1^2 \quad ==> \quad (n+1)^2 = 121$ $n + 1 = \sqrt{121} = 11 \quad ==> \quad n = 10$ OR $3 \times 10^2 + (6 \times 10) = 360$ $\therefore$ Number of terms = 10	1 1 1 1
	SECTION E	
20A	Join OA , (i) $\angle OAB = \angle OBA = 30^0$ $\quad [OA = OB]$ $\angle OAC = \angle OCA = 20^0$ $\quad [OA = OC]$ $\angle BAC = \angle OAB + \angle OAC = 30 + 20 = 50^0$ (ii) $\angle BOC = 2 \times 50 = 100^0$ $\quad [\angle BOC = 2 \times \angle BAC]$ 	1 1 1 1
20B	(i) $\angle OAC = \frac{x^0}{2}$ $\quad [\angle BAC = x^0]$ (ii) $\angle OAC + \angle OCA = \frac{180 - 50}{2} = 65^0$ $\angle AOC = 180 - \frac{130}{2} = 180 - 65 = 115^0$ 	1 1 1

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21	(i) $\angle BCD = 180 - 80 = 100^0$ [Sum of the angles of a cyclic quadrilateral is 180^0] 	1
	$\angle CBE = 180 - 70 = 110^0$	
	$[\ \angle ABC = 180 - 110 = 70^0\ ]$	1
	(ii) Take $\angle CBE = x^0$	
	Then $\angle ABC = 180 - x^0$	1
	$==>\quad \angle ADC = x^0$	1
22A	(i) AS = AP = 3 cm 	1
	(ii) BP = BQ	
	CQ = CR	
	DR = DS	1
	AB + CD = AP + BP + CR + DR	
	= AS + BQ + CQ + DS	1
	= (AS + DS) + (BQ + CQ)	
	= AD + BC	1
22B	It is aright triangle.	1
	$2r = 6 + 8 - 10 \quad ==> \quad 2r = 4$	1
	$r = 2\ cm$	1
	OR	
	It is aright triangle.	
	Area of the triangle $= \frac{1}{2} \times 6 \times 8 = 24\ sq.cm$	
	Perimeter of the triangle $= 6 + 8 + 10 = 24\ cm$	
	$r = \frac{A}{S} = \frac{24}{12} = 2\ cm$	
23	(i) $\angle BAC = 90^0$ 	1
	(ii) $r = \frac{5}{2 \times 0.6428}\ cm$	1
	$[\ \sin 40^0 = \frac{5}{2r}$ OR $\cos 50^0 = \frac{5}{2r}\ ]$	
	(iii) $AB = 5 \times 1.1918\ cm$.	1
	$\left[\ \tan 50^0 = \frac{AB}{5}\ \right]$	
	Perimeter of the triangle $= 5 + (5 \times 1.1918) + \frac{5}{0.6428}\ cm$	1
24A	(i) 	1

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(ii) In larger right triangle , $\tan 40^0 = \frac{30-x}{x} \implies 0.8391 = \frac{30-x}{x}$ — 1

$0.8391x = 30 - x \implies 1.8391x = 30 \implies x = \frac{30}{1.8391}$

Height of the post $= \frac{30}{1.8391}\ m$ — 1

OR

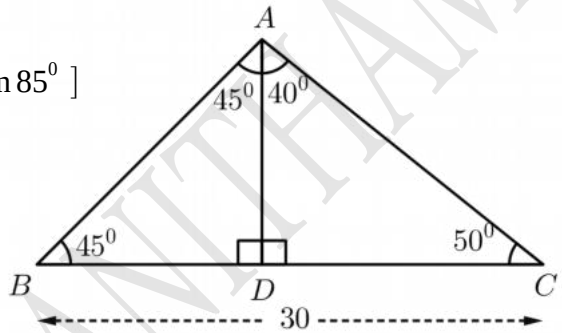
In larger right triangle , $\tan 50^0 = \frac{x}{30-x} \implies 1.1918 = \frac{x}{30-x}$ — 1

$1.1918(30-x) = x \implies 2.1918x = 1.1918 \times 30 \implies x = \frac{1.1918 \times 30}{2.1918}$

Height of the post $= \frac{1.1918 \times 30}{2.1918}\ m$

OR

$2r = \frac{30}{\sin 85^0} = \frac{30}{0.9962}$ $\quad [\ BC = 2r \sin 85^0\ ]$



$AB = 2r \sin 50^0 = \frac{30}{0.9962} \times 0.7660\ m$

$AD = \frac{1}{\sqrt{2}} \times AB$

$= \frac{1}{\sqrt{2}} \times \frac{30}{0.9962} \times 0.7660\ m$

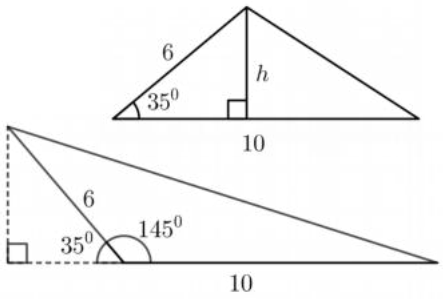
24B

(i) $h = 6 \times \sin 35^0 = 6 \times 0.5736\ cm$

Area of the triangle $= \frac{1}{2} \times 10 \times 6 \times 0.5736$

$= 30 \times 0.5736\ sq.cm$

(ii) Area of the triangle $= 30 \times 0.5736\ sq.cm$



25A

(i) Slant height $= \sqrt{25^2 - 15^2} = 20\ cm$ $\quad \left[\ \left(\frac{a}{2}\right)^2 + l^2 = e^2\ \right]$ — 1

(ii) Height $= \sqrt{20^2 - 15^2} = \sqrt{175} = 5\sqrt{7}\ cm$ $\quad \left[\ \left(\frac{a}{2}\right)^2 + h^2 = l^2\ \right]$ — 2

(iii) Volume $= \frac{1}{3} \times 30^2 \times 5\sqrt{7} = 1500\sqrt{7}\ cubic.cm$ $\quad \left[\ V = \frac{1}{3}a^2h\ \right]$ — 2

25B

(i) Slant height $= 25\ cm$ — 1

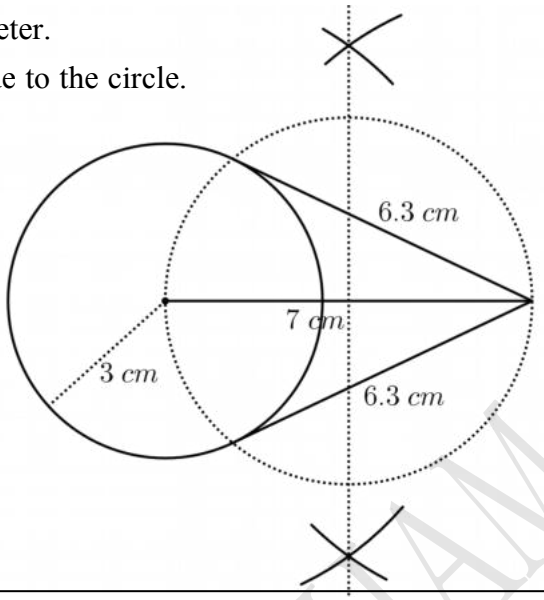
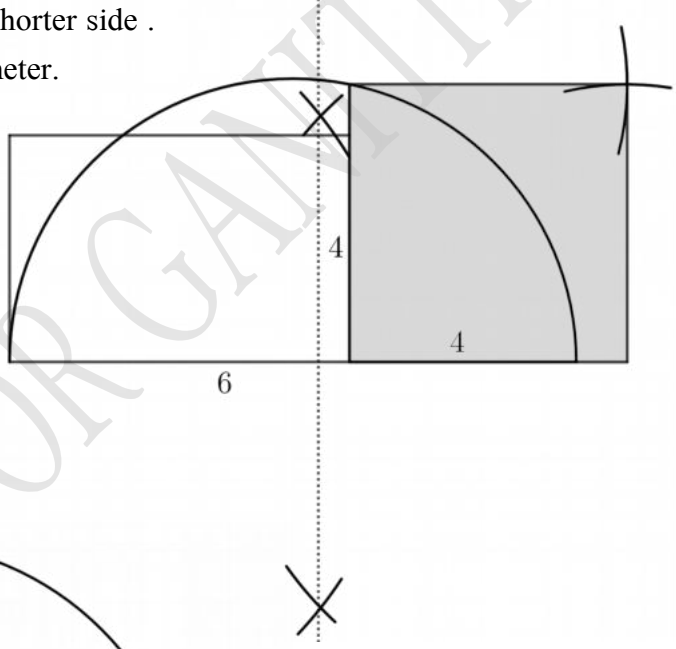
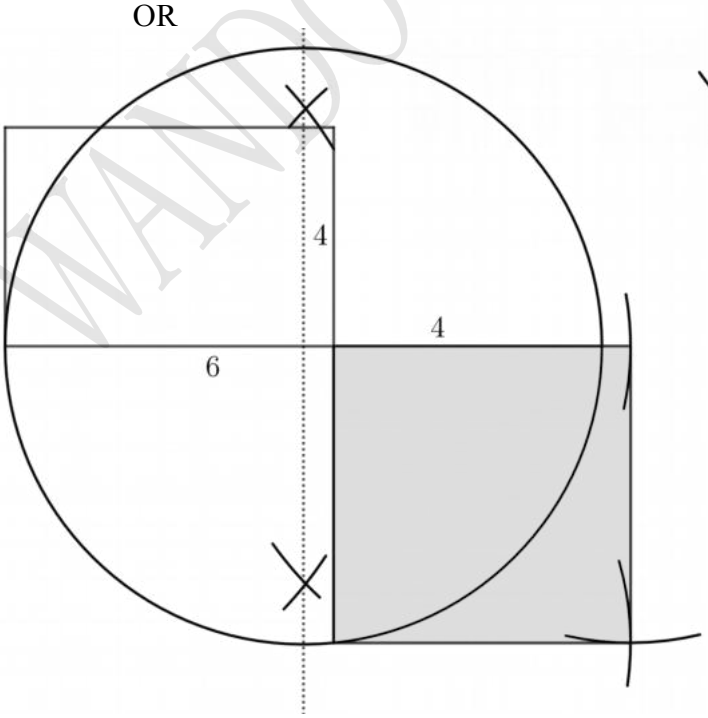
(ii) $r = \frac{216}{360} \times 25 = 15\ cm$ $\quad \left[\ \frac{x}{360} = \frac{r}{R}\ \right]$ — 2

(iii) Height $= \sqrt{25^2 - 15^2} = 20\ cm$ $\quad [\ r^2 + h^2 = l^2\ ]$ — 1

Volume $= \frac{1}{3} \times \pi \times 15^2 \times 20 = 1500\pi\ cubic.cm$ $\quad \left[\ V = \frac{1}{3}a^2h\ \right]$ — 1

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26	For drawing a circle of radius 3 cm and mark point 7 cm away from the centre. For drawing a circle with this line as diameter. For drawing tangents from the point outside to the circle. For measuring the tangents. 	1 1 1 1
27	For drawing the rectangle. For extending the longer side by the shorter side . For drawing a circle with line as diameter. For completing the square.  OR 	1 1 1 1 1

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